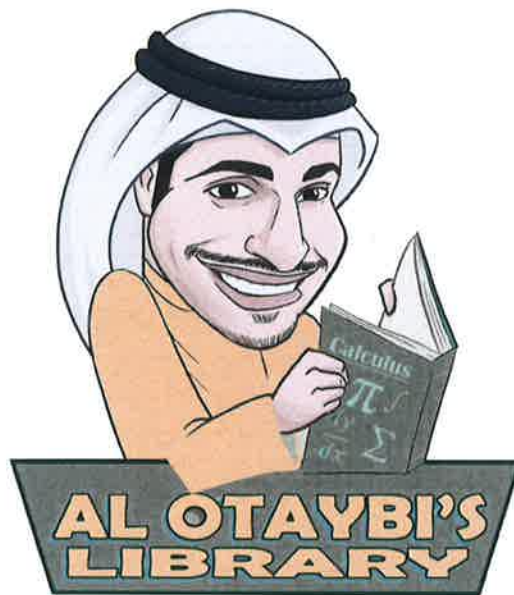


Kuwait University

2021/2022

Spring course



Calculus B

Second

Section : 11.1 (Part b/d)

11.1 Sequences.

(8 b)



11.1 Sequences



المتتالية هي مجموعة من الحدود تربط بينهم علاقة

$$\{a_n\} = \{a_1, a_2, a_3, \dots\}$$

1st term

2nd term

3rd term

مثال توضيحي

$$\left\{ \frac{n}{n+1} \right\} = \left[\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots \right]$$

1st term a_1 2nd term a_2 3rd term a_3 4th term a_4

التعويض يكون بأعداد صحيحة موجبة

في أي حال من الأحوال لا يمكن اشتقاقها أو تطبيق اللوبيتال الا بعد اعادة تسميتها



جريمة



a_{-2}

ما يجبر أكون بعدد $\ominus$

$a_{\frac{7}{2}}$

ما يجبر أكون بكس

Find the first four terms of the following sequences

$$a_n = \frac{n}{n^2 + 1}$$

$$\underline{n=1} \rightarrow a_1 = \frac{1}{(1)^2 + 1} \Rightarrow a_1 = \frac{1}{2}$$

$$\underline{n=2} \rightarrow a_2 = \frac{2}{(2)^2 + 1} = \frac{2}{5} \Rightarrow a_2 = \frac{2}{5}$$

$$\underline{n=3} \rightarrow a_3 = \frac{3}{(3)^2 + 1} = \frac{3}{9+1} = \frac{3}{10} \Rightarrow a_3 = \frac{3}{10}$$

$$\underline{n=4} \rightarrow a_4 = \frac{4}{(4)^2 + 1} = \frac{4}{16+1} = \frac{4}{17} \Rightarrow a_4 = \frac{4}{17}$$

$$a_1 = 1, \quad a_{n+1} = \frac{a_n}{n+3} \quad n \geq 1$$

$$a_1 = 1$$

$$\underline{n=1} \rightarrow a_2 = \frac{a_1}{1+3} = \frac{1}{4} \Rightarrow a_2 = \frac{1}{4}$$

$$\underline{n=2} \rightarrow a_3 = \frac{a_2}{2+3} = \frac{\frac{1}{4}}{5} = \frac{1}{20} \Rightarrow a_3 = \frac{1}{20}$$

$$\underline{n=3} \rightarrow a_4 = \frac{a_3}{3+3} = \frac{\frac{1}{20}}{6} = \frac{1}{120} \Rightarrow a_4 = \frac{1}{120}$$



Text book question



List the first five terms of the sequence.

$$a_n = \frac{3(-1)^n}{n!}$$

$$n=1 \rightarrow a_1 = \frac{3(-1)^1}{1!} = \frac{-3}{1} = -3$$

$$n=2 \rightarrow a_2 = \frac{3(-1)^2}{2!} = \frac{3}{2!} = \frac{3}{2}$$

$$n=3 \rightarrow a_3 = \frac{3(-1)^3}{3!} = \frac{-3}{3!}$$

$$n=4 \rightarrow a_4 = \frac{3(-1)^4}{4!} = \frac{3}{4!}$$

$$n=5 \rightarrow a_5 = \frac{3(-1)^5}{5!} = \frac{-3}{5!}$$



Text book question



List the first five terms of the sequence.

$$a_1 = 2, \quad a_{n+1} = \frac{a_n}{1 + a_n}$$

$$a_1 = 2$$

$$\underline{n=1} \rightarrow a_2 = \frac{a_1}{1+a_1} = \frac{2}{1+2} = \frac{2}{3}$$

$$\underline{n=2} \rightarrow a_3 = \frac{a_2}{1+a_2} = \frac{\frac{2}{3}}{1+\frac{2}{3}} = \frac{\frac{2}{3}}{\frac{3+2}{3}} = \frac{2}{5}$$

$$\underline{n=3} \rightarrow a_4 = \frac{a_3}{1+a_3} = \frac{\frac{2}{5}}{1+\frac{2}{5}} = \frac{\frac{2}{5}}{\frac{5+2}{5}} = \frac{2}{7}$$

$$\underline{n=4} \rightarrow a_5 = \frac{a_4}{1+a_4} = \frac{\frac{2}{7}}{1+\frac{2}{7}} = \frac{\frac{2}{7}}{\frac{7+2}{7}} = \frac{2}{9}$$

Determine whether the sequence $\{a_n\}$ is convergent or divergent . if converges find the limit

$$a_n = \left[\left(1 + \frac{3}{2n+1} \right)^{5n} - \left(\frac{3}{\pi} \right)^n \right]$$

$$\lim_{n \rightarrow \infty} \left[\left(1 + \frac{3}{2n+1} \right)^{5n} - \left(\frac{3}{\pi} \right)^n \right]$$

$$= e^{\frac{3(5)}{2}} - 0 = e^{\frac{15}{2}}$$

since $|\frac{3}{\pi}| < 1$

$\Rightarrow \therefore a_n$ is converges to $e^{\frac{15}{2}}$

$$a_n = \frac{2\ln(n)}{\ln(2n)}$$

$$\lim_{n \rightarrow \infty} \frac{2\ln n}{\ln(2n)} \quad \frac{\infty}{\infty}$$

Let $f(x) = \frac{2\ln x}{\ln(2x)}$

$$\lim_{x \rightarrow \infty} \frac{2\ln x}{\ln(2x)} \quad \frac{\infty}{\infty}$$

L'HOP

$$\lim_{x \rightarrow \infty} \frac{2 \cdot \frac{1}{x}}{\frac{2}{2x}} = \lim_{x \rightarrow \infty} (2) = 2$$

$\Rightarrow \therefore a_n$ is converges to 2



Text book questions



**Determine whether the sequence is converges or diverges.
If converges find the limit.**

$$a_n = \frac{\cos^2 n}{2^n}$$

$$\lim_{n \rightarrow \infty} \frac{\cos^2 n}{2^n}$$

$$\div 2^n \quad 0 \leq \cos^2 n \leq 1$$

$$0 \leq \frac{\cos^2 n}{2^n} \leq \frac{1}{2^n}$$

$$\lim_{n \rightarrow \infty} 0 = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{2^n} = \frac{1}{\infty} = 0$$

∴ By Sandwich Theorem $\rightarrow \lim_{n \rightarrow \infty} \frac{\cos^2 n}{2^n} = 0 \rightarrow \therefore a_n$ is converges

**Determine whether the sequence is converges or diverges.
If converges find the limit.**

$$a_n = n \sin\left(\frac{1}{n}\right)$$

$$\lim_{n \rightarrow \infty} n \sin\left(\frac{1}{n}\right)$$

$$\lim_{n \rightarrow \infty} \cancel{n} \times \cancel{\frac{1}{n}} \frac{\sin(\frac{1}{n})}{\frac{1}{n}} = 1$$

∴ a_n is convergent.



Text book questions



Determine whether the sequence is converges or diverges.
If converges find the limit.

$$a_n = \frac{3 + 5n^2}{n + n^2}$$

$$\lim_{n \rightarrow \infty} \frac{3 + 5n^2}{n + n^2}$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{n^2} \left[\frac{3}{n^2} + 5 \right]}{\cancel{n^2} \left[\frac{1}{n} + 1 \right]} = \frac{0 + 5}{0 + 1} = 5$$

$\therefore a_n$ is convergent

Determine whether the sequence is converges or diverges.
If converges find the limit.

$$a_n = \sqrt{\frac{n+1}{9n+1}}$$

$$\lim_{n \rightarrow \infty} \sqrt{\frac{n+1}{9n+1}}$$

$$\lim_{n \rightarrow \infty} \sqrt{\frac{\cancel{n} \left[1 + \frac{1}{n} \right]}{\cancel{n} \left[9 + \frac{1}{n} \right]}} = \sqrt{\frac{1+0}{9+0}} = \sqrt{\frac{1}{9}} = \frac{1}{3}$$

$\therefore a_n$ is convergent.

Determine whether the sequence $\{a_n\}$ is convergent or divergent . if converges find the limit

$$a_n = n [\ln(n+1) - \ln(n)]$$

$$\lim_{n \rightarrow \infty} n [\ln(n+1) - \ln(n)]$$

$$\lim_{n \rightarrow \infty} n \ln \left(\frac{n+1}{n} \right)$$

$$\lim_{n \rightarrow \infty} \ln \left(\frac{n+1}{n} \right)^n = \lim_{n \rightarrow \infty} \ln \left(\frac{n}{n} + \frac{1}{n} \right)^n$$

$$\lim_{n \rightarrow \infty} \ln \left(1 + \frac{1}{n} \right)^n = \ln e^1 = 1$$

$\Rightarrow \therefore a_n$ is convergent

$$a_n = \frac{n^2}{3n+5} \sin \left(\frac{1}{n} \right)$$

$$\lim_{n \rightarrow \infty} \frac{n^2}{3n+5} \sin \left(\frac{1}{n} \right)$$

$$\lim_{n \rightarrow \infty} \frac{n^2}{n(3+\frac{5}{n})} \sin \left(\frac{1}{n} \right)$$

$$\frac{\infty}{\infty} \sin 0$$

$$\lim_{n \rightarrow \infty} \frac{n}{3+\frac{5}{n}} \frac{\sin \left(\frac{1}{n} \right)}{\frac{1}{n}} = \frac{1}{3+0} (1) = \frac{1}{3}$$

$\Rightarrow \therefore a_n$ is convergent

Determine whether the sequence $\{a_n\}$ is convergent or divergent . if converges find the limit

$$a_n = \frac{n + \cos n}{\sqrt{n^2 + 1}}$$

$$\lim_{n \rightarrow \infty} \frac{n + \cos n}{\sqrt{n^2 + 1}}$$

$$\lim_{n \rightarrow \infty} \left[\frac{n}{\sqrt{n^2 + 1}} + \frac{\cos n}{\sqrt{n^2 + 1}} \right]$$

$$\lim_{n \rightarrow \infty} \frac{n}{\sqrt{n^2 + 1}}$$

$$\lim_{n \rightarrow \infty} \frac{n}{\sqrt{n^2(1 + \frac{1}{n^2})}}$$

$$\lim_{n \rightarrow \infty} \frac{n}{n\sqrt{1 + \frac{1}{n^2}}}$$

$$= \frac{1}{\sqrt{1 + 0}} = 1$$

$$+ \lim_{n \rightarrow \infty} \frac{\cos n}{\sqrt{n^2 + 1}}$$

$$-1 \leq \cos n \leq 1$$

$$\div \sqrt{n^2 + 1}$$

$$\frac{-1}{\sqrt{n^2 + 1}} \leq \frac{\cos n}{\sqrt{n^2 + 1}} \leq \frac{1}{\sqrt{n^2 + 1}}$$

$$\lim_{n \rightarrow \infty} \frac{-1}{\sqrt{n^2 + 1}} = \frac{-1}{\infty} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n^2 + 1}} = \frac{1}{\infty} = 0$$

∴ By Sandwich Theorem

$$\lim_{n \rightarrow \infty} \frac{\cos n}{\sqrt{n^2 + 1}} = 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{n + \cos n}{\sqrt{n^2 + 1}} = 1 + 0 = 1$$

∴ a_n is Convergent

$$a_n = \frac{(-1)^n \sin n}{n^2 + 1}$$

$$\lim_{n \rightarrow \infty} |a_n|$$

$$\lim_{n \rightarrow \infty} \frac{|\sin n|}{n^2 + 1}$$

$$0 \leq |\sin n| \leq 1$$

$$\div (n^2 + 1)$$

$$0 \leq \frac{|\sin n|}{n^2 + 1} \leq \frac{1}{n^2 + 1}$$

$$\lim_{n \rightarrow \infty} 0 = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n^2 + 1} = \frac{1}{\infty} = 0$$

∴ By Sandwich Thm

$$\lim_{n \rightarrow \infty} |a_n| = 0$$

$$\therefore \lim_{n \rightarrow \infty} a_n = 0$$

∴ a_n is converges to 0

Determine whether the sequence $\{a_n\}$ is convergent or divergent. If converges find the limit

$$a_n = \left\{ n \tan\left(\frac{\pi}{n}\right) + \frac{2n^2}{n^2 + n} \tan^{-1} n \right\}$$

$$\lim_{n \rightarrow \infty} \left[n \tan\left(\frac{\pi}{n}\right) + \frac{2n^2}{n^2 + n} \tan^{-1} n \right]$$

$$\lim_{n \rightarrow \infty} n \tan\left(\frac{\pi}{n}\right) + \lim_{n \rightarrow \infty} \frac{2n^2}{n^2 + n} \tan^{-1} n$$

$\infty \cdot 0$

$$\lim_{n \rightarrow \infty} n \cdot \frac{\tan\left(\frac{\pi}{n}\right)}{\frac{\pi}{n}} \cdot \frac{\pi}{n}$$

$$= (1) \pi = \pi$$

$$\lim_{n \rightarrow \infty} \frac{2n^2}{n^2(1 + \frac{1}{n})} \tan^{-1} n$$

$$= \frac{2}{1 + 0} \frac{\pi}{2}$$

$$= 2 \frac{\pi}{2} = \pi$$

$$\Rightarrow \therefore \lim_{n \rightarrow \infty} a_n = \pi + \pi = 2\pi$$

$\Rightarrow \therefore a_n$ is convergent $\triangle$

Determine whether the sequence $\{a_n\}$ is convergent or divergent . if converges find the limit

$$a_n = \frac{5 + 3^n - 2^n}{3 + 3^n}$$

$$\lim_{n \rightarrow \infty} \frac{5 + 3^n - 2^n}{3 + 3^n} \quad \frac{\infty}{\infty}$$

$$\lim_{n \rightarrow \infty} \frac{5(\frac{1}{3})^n + 1 - (\frac{2}{3})^n}{3(\frac{1}{3})^n + 1}$$

$$= \frac{0 + 1 - 0}{0 + 1} = \frac{1}{1} = 1$$

$\Rightarrow \therefore a_n$ is convergent

$$a_n = n(\sqrt[n]{23} - 1)$$

$$\lim_{n \rightarrow \infty} n[(23)^{\frac{1}{n}} - 1] \quad \begin{matrix} \infty [23^0 - 1] \\ \infty [1 - 1] \\ \infty \cdot 0 \end{matrix}$$

$$\text{Let } f(x) = x[(23)^{\frac{1}{x}} - 1]$$

$$\lim_{x \rightarrow \infty} \frac{(23)^{\frac{1}{x}} - 1}{\frac{1}{x}} \quad \frac{0}{0}$$

$$\text{L'Hop} \quad \lim_{x \rightarrow \infty} \frac{(23)^{\frac{1}{x}} \left(-\frac{1}{x^2}\right) \ln(23)}{-\frac{1}{x^2}}$$

$$= (23)^0 \ln(23) = \ln 23$$

$\Rightarrow \therefore a_n$ is convergent

Determine whether the sequence $\{a_n\}$ is convergent or divergent. if converges find the limit

$$a_n = \left\{ \frac{3^n}{2^n - 1} - \frac{3^n}{2^n + 1} \right\}$$

$$\lim_{n \rightarrow \infty} \left[\frac{3^n}{2^n - 1} - \frac{3^n}{2^n + 1} \right] \quad \infty - \infty$$

$$\lim_{n \rightarrow \infty} \left[\frac{3^n (2^n + 1)}{(2^n - 1)(2^n + 1)} - \frac{3^n (2^n - 1)}{(2^n + 1)(2^n - 1)} \right]$$

$$\lim_{n \rightarrow \infty} \frac{3^n (2^n + 1) - 3^n (2^n - 1)}{(2^n - 1)(2^n + 1)}$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{6^n} + 3^n - \cancel{6^n} + 3^n}{4^n - 1}$$

$$\lim_{n \rightarrow \infty} \frac{2(3^n)}{4^n - 1}$$

$$\div 4^n \quad \lim_{n \rightarrow \infty} \frac{2 \frac{3^n}{4^n}}{\frac{4^n}{4^n} - \frac{1}{4^n}}$$

$$\lim_{n \rightarrow \infty} \frac{2 \left(\frac{3}{4} \right)^n}{1 - \left(\frac{1}{4} \right)^n} = \frac{0}{1 - 0} = 0$$

$\Rightarrow \therefore a_n$ is convergent

$$\lim_{n \rightarrow \infty} \frac{3^n}{2^n - 1} - \lim_{n \rightarrow \infty} \frac{3^n}{2^n + 1}$$

$$\div 2^n \quad \lim_{n \rightarrow \infty} \frac{\left(\frac{3}{2} \right)^n}{1 - \left(\frac{1}{2} \right)^n} - \lim_{n \rightarrow \infty} \frac{\left(\frac{3}{2} \right)^n}{1 + \left(\frac{1}{2} \right)^n}$$

$$\frac{\infty}{1 - 0} - \frac{\infty}{1 + 0}$$

$$\infty - \infty$$

توحيد مقامات

$$\left(\frac{3}{4} \right)^{\infty} = 0$$

Determine whether the sequence $\{a_n\}$ is convergent or divergent. If converges find the limit

$$a_n = 2ne^{-n} + 3n \tan\left(\frac{1}{n}\right)$$

$$\lim_{n \rightarrow \infty} \left[2ne^{-n} + 3n \tan\left(\frac{1}{n}\right) \right]$$

$$\lim_{n \rightarrow \infty} 2ne^{-n}$$

$\infty \cdot 0$

Let $f(x) = 2xe^{-x}$

$$\lim_{x \rightarrow \infty} \frac{2x}{e^x} \quad \frac{\infty}{\infty}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{2}{e^x}$$

$$= \frac{2}{\infty} = 0$$

$$+ \lim_{n \rightarrow \infty} 3n \tan\left(\frac{1}{n}\right)$$

$\infty \cdot 0$

$$\lim_{n \rightarrow \infty} 3n \left(\frac{\tan\left(\frac{1}{n}\right)}{\frac{1}{n}} \right) \quad \cancel{\frac{1}{n}}$$

$$= 3(1)$$

$$= 3$$

$$\Rightarrow \therefore \lim_{n \rightarrow \infty} a_n = 0 + 3 = 3$$

$$\Rightarrow \therefore a_n \text{ is convergent}$$

Determine whether the sequence $\{a_n\}$ is convergent or divergent. If converges find the limit

$$a_n = \frac{(-1)^n + \sin(n)}{n+1}$$

$$\lim_{n \rightarrow \infty} \frac{(-1)^n + \sin n}{n+1}$$

$$\lim_{n \rightarrow \infty} \left[\frac{(-1)^n}{n+1} + \frac{\sin n}{n+1} \right]$$

$$\lim_{n \rightarrow \infty} \frac{(-1)^n}{n+1} + \lim_{n \rightarrow \infty} \frac{\sin n}{n+1}$$

$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^n}{n+1} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{1}{n+1} \right|$$

$$= \left| \frac{1}{\infty} \right|$$

$$= 0$$

$$\Rightarrow \therefore \lim_{n \rightarrow \infty} \frac{(-1)^n}{n+1} = 0$$

$$-1 \leq \sin n \leq 1$$

$$\div (n+1)$$

$$\frac{-1}{n+1} \leq \frac{\sin n}{n+1} \leq \frac{1}{n+1}$$

$$\lim_{n \rightarrow \infty} \frac{-1}{n+1} = \frac{-1}{\infty} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n+1} = \frac{1}{\infty} = 0$$

$\therefore$ By Sandwich Theorem

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{\sin n}{n+1} = 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} a_n = 0 + 0 = 0$$

$\Rightarrow \therefore a_n$ is convergent

Determine whether the sequence $\{b_n\}$ is convergent or divergent. If converges find the limit

$$b_n = \{\sqrt[n]{2} + (-1)^n n e^{-n}\}$$

$$\lim_{n \rightarrow \infty} [\sqrt[n]{2} + (-1)^n n e^{-n}]$$

$$\lim_{n \rightarrow \infty} (2)^{\frac{1}{n}} + \lim_{n \rightarrow \infty} (-1)^n n e^{-n}$$

$$= 2^{\frac{1}{\infty}}$$

$$= 2^0$$

$$= 1$$

$$\lim_{n \rightarrow \infty} |(-1)^n n e^{-n}|$$

$$\lim_{n \rightarrow \infty} |n e^{-n}|$$

$$\text{Let } f(x) = x e^{-x}$$

$$\lim_{x \rightarrow \infty} \left| \frac{x}{e^x} \right|$$

L'HOP

$$\lim_{x \rightarrow \infty} \left| \frac{1}{e^x} \right| = \left| \frac{1}{\infty} \right| = 0$$

$$\therefore \lim_{n \rightarrow \infty} (-1)^n n e^{-n} = 0$$

$$\Rightarrow \therefore \lim_{n \rightarrow \infty} b_n = 1 + 0 = 1$$

$\Rightarrow \therefore b_n$ is convergent

Determine whether the sequence $\{b_n\}$ is convergent or divergent. If converges find the limit

$$b_n = n^2 \left(\sqrt{\frac{n}{n+1}} - \sqrt{\frac{n+2}{n+3}} \right)$$

$$\lim_{n \rightarrow \infty} n^2 \left(\sqrt{\frac{n}{n+1}} - \sqrt{\frac{n+2}{n+3}} \right) \quad \begin{matrix} \infty(1-1) \\ \infty \cdot 0 \end{matrix} \quad \frac{\sqrt{\frac{n}{n+1}} + \sqrt{\frac{n+2}{n+3}}}{\sqrt{\frac{n}{n+1}} + \sqrt{\frac{n+2}{n+3}}}$$

$$\lim_{n \rightarrow \infty} \frac{n^2 \left(\frac{n}{n+1} - \frac{n+2}{n+3} \right)}{\sqrt{\frac{n}{n+1}} + \sqrt{\frac{n+2}{n+3}}} \quad \text{مليق}$$

$$\lim_{n \rightarrow \infty} \frac{n^2 \left(\frac{n(n+3) - (n+2)(n+1)}{(n+1)(n+3)} \right)}{\sqrt{\frac{n}{n+1}} + \sqrt{\frac{n+2}{n+3}}} \rightarrow \frac{n^2 + 3n - (n^2 + n + 2n + 2)}{n^2 + 3n - n^2 - 3n - 2}$$

$$\lim_{n \rightarrow \infty} \frac{n^2 (-2)}{(n+1)(n+3) \left(\sqrt{\frac{n}{n+1}} + \sqrt{\frac{n+2}{n+3}} \right)}$$

$$\lim_{n \rightarrow \infty} \frac{-2n^2}{n(1 + \frac{1}{n}) n(1 + \frac{3}{n}) \left(\sqrt{\frac{n}{n+1}} + \sqrt{\frac{n+2}{n+3}} \right)}$$

$$= \frac{-2}{(1+0)(1+0)(\sqrt{1} + \sqrt{1})} = \frac{-2}{(1)(1)(1+1)} = \frac{-2}{2} = -1$$

$\Rightarrow \therefore b_n$ is convergent

Determine whether the sequence $\{a_n\}$ is convergent or divergent . if converges find the limit

$$a_n = (2n + 1)(n - 1)! \sin\left(\frac{1}{n!}\right)$$

$$\lim_{n \rightarrow \infty} (2n+1)(n-1)! \sin\left(\frac{1}{n!}\right) \quad \infty \sin 0$$

$$\lim_{n \rightarrow \infty} (2n+1)(n-1)! \frac{\sin\left(\frac{1}{n!}\right)}{\frac{1}{n!}}$$

$$\lim_{n \rightarrow \infty} \frac{(2n+1)(n-1)!}{n!} \frac{\sin\left(\frac{1}{n!}\right)}{\frac{1}{n!}}$$

$$\lim_{n \rightarrow \infty} \frac{(2n+1) \cancel{(n-1)!}}{n \cancel{(n-1)!}} \frac{\sin\left(\frac{1}{n!}\right)}{\frac{1}{n!}}$$

$$\lim_{n \rightarrow \infty} \frac{2n+1}{n} \frac{\sin\left(\frac{1}{n!}\right)}{\frac{1}{n!}}$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{n} \left(2 + \frac{1}{n}\right)}{\cancel{n}} \frac{\sin\left(\frac{1}{n!}\right)}{\frac{1}{n!}}$$

$$= (2+0)(1) = 2$$

$\Rightarrow \therefore a_n$ is Convergent

Determine whether the sequence $\{a_n\}$ is convergent or divergent . if converges find the limit

$$a_n = \frac{n!}{n^n} + n \sin\left(\frac{\pi}{2n}\right)$$

$$\lim_{n \rightarrow \infty} \left[\frac{n!}{n^n} + n \sin\left(\frac{\pi}{2n}\right) \right]$$

$$\lim_{n \rightarrow \infty} \frac{n!}{n^n}$$

+

$$\lim_{n \rightarrow \infty} n \sin\left(\frac{\pi}{2n}\right)$$

$$\frac{n!}{n^n} = \frac{n(n-1)(n-2) \dots (3)(2)(1)}{n \cdot n \cdot n \dots n \cdot n \cdot n}$$

$$0 < \frac{n!}{n^n} < \frac{2}{n}$$

$$\lim_{n \rightarrow \infty} 0 = 0$$

$$\lim_{n \rightarrow \infty} \frac{2}{n} = \frac{2}{\infty} = 0$$

$\therefore$ By Sandwich Theorem

$$\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$$

$$\begin{aligned} \lim_{n \rightarrow \infty} n \sin\left(\frac{\pi}{2n}\right) &= \lim_{n \rightarrow \infty} \frac{\sin\left(\frac{\pi}{2n}\right)}{\frac{\pi}{2n}} \\ &= (1) \frac{\pi}{2} \\ &= \frac{\pi}{2} \end{aligned}$$

$$\Rightarrow \therefore \lim_{n \rightarrow \infty} a_n = 0 + \frac{\pi}{2} = \frac{\pi}{2}$$

$\therefore a_n$ is convergent

Determine whether the sequence

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$$a_n = \frac{3^n}{n!}$$

converges or diverges. If converges, find the limit.

$$\frac{3^n}{n!} = \frac{(3)(3)(3) \dots (3)(3)(3)}{n(n-1)(n-2) \dots (3)(2)(1)}$$

$$0 < \frac{3^n}{n!} < \frac{3}{n} \cdot \frac{9}{2}$$

$$\lim_{n \rightarrow \infty} 0 = 0$$

$$\lim_{n \rightarrow \infty} \frac{3}{n} \cdot \frac{9}{2} = 0$$

$\therefore$ By Sandwich Theorem

$$\lim_{n \rightarrow \infty} \frac{3^n}{n!} = 0$$

$\therefore a_n$ is convergent

Determine whether the sequence $\{a_n\}$ is convergent or divergent . if converges find the limit

$$a_n = \frac{(-3)^n}{n!}$$

$$\lim_{n \rightarrow \infty} \frac{(-1)^n 3^n}{n!}$$

$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^n 3^n}{n!} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{3^n}{n!} \right|$$

$$\frac{3^n}{n!} = \frac{(3)(3)(3) \dots \cancel{(3)(3)(3)}}{n(n-1)(n-2) \dots \cancel{(3)(2)(1)}}$$

$$0 < \frac{3^n}{n!} < \frac{3}{n} \quad \frac{3(3)}{2}$$

$$\lim_{n \rightarrow \infty} 0 = 0$$

$$\lim_{n \rightarrow \infty} \frac{3}{n} = \frac{3(3)}{2} = 0$$

$$\Rightarrow \therefore \text{By Sandwich Theorem}$$
$$\lim_{n \rightarrow \infty} \frac{3^n}{n!} = 0$$

$$\Rightarrow \therefore \lim_{n \rightarrow \infty} \frac{(-1)^n 3^n}{n!} = 0$$

$\Rightarrow \therefore a_n$ is Convergent

Determine whether the sequence $\{a_n\}$ is convergent or divergent . if converges find the limit

$$a_n = \frac{n!}{2^n}$$

$$\frac{2^n}{n!} = \frac{(2)(2)(2) \cdots (2)(\cancel{2})(2)}{n(n-1)(n-2) \cdots (3)(\cancel{2})(1)}$$

$$0 < \frac{2^n}{n!} < \frac{2}{n} \cdot 2$$

$$\lim_{n \rightarrow \infty} 0 = 0$$

$$\lim_{n \rightarrow \infty} \frac{2}{n} \cdot 2 = \frac{4}{\infty} = 0$$

∴ By Sandwich Theorem

$$\lim_{n \rightarrow \infty} \frac{2^n}{n!} = 0$$

$$\Rightarrow \therefore \lim_{n \rightarrow \infty} \frac{n!}{2^n} = \frac{1}{0} = \infty$$

$\Rightarrow \therefore$ Ans is Divergent.

List the first three terms of the sequence

Fall 2018

$$a_1 = 1, \quad a_n = \frac{3\sqrt{n}}{\sqrt{n} + 2}, \quad n = 1, 2, \dots$$

and determine whether it is convergent or divergent, and find its limit if possible.

$$a_1 = 1$$

$$\underline{n=2} \longrightarrow a_2 = \frac{3\sqrt{2}}{\sqrt{2} + 2}$$

$$\underline{n=3} \longrightarrow a_3 = \frac{3\sqrt{3}}{\sqrt{3} + 2}$$

$$\lim_{n \rightarrow \infty} \frac{3\sqrt{n}}{\sqrt{n} + 2}$$

$$\lim_{n \rightarrow \infty} \frac{3\cancel{\sqrt{n}}}{\cancel{\sqrt{n}} \left[1 + \frac{2}{\sqrt{n}} \right]} = \frac{3}{1+0} = 3$$

$\Rightarrow \therefore a_n$ is convergent.



True or False (Justify)



The sequence

$$a_n = \sqrt[n]{2^n + 3^n + 4^n}$$

is divergent.

Past exam

$$\lim_{n \rightarrow \infty} (2^n + 3^n + 4^n)^{\frac{1}{n}} \quad \infty^0$$

$$y = (2^x + 3^x + 4^x)^{\frac{1}{x}}$$

$$\ln y = \ln (2^x + 3^x + 4^x)^{\frac{1}{x}}$$

$$\ln y = \frac{1}{x} \ln (2^x + 3^x + 4^x)$$

$$\lim_{x \rightarrow \infty} \frac{\ln(2^x + 3^x + 4^x)}{x} \quad \frac{\ln \infty}{\infty} \quad \frac{\infty}{\infty}$$

L'Hop

$$\lim_{x \rightarrow \infty}$$

$$\frac{\frac{2^x \ln 2 + 3^x \ln 3 + 4^x \ln 4}{2^x + 3^x + 4^x}}{1}$$

$$\lim_{x \rightarrow \infty}$$

$$\frac{2^x \ln 2 + 3^x \ln 3 + 4^x \ln 4}{2^x + 3^x + 4^x}$$

$\div 4^x$

$$\lim_{x \rightarrow \infty}$$

$$\frac{\left(\frac{2}{4}\right)^x \ln 2 + \left(\frac{3}{4}\right)^x \ln 3 + \ln 4}{\left(\frac{2}{4}\right)^x + \left(\frac{3}{4}\right)^x + 1}$$

$$= \frac{0 + 0 + \ln 4}{0 + 0 + 1} = \ln 4$$

$$\Rightarrow \therefore \lim_{x \rightarrow \infty} (2^x + 3^x + 4^x)^{\frac{1}{x}} = e^{\ln 4} = 4$$

$\therefore a_n$ is convergent $\rightarrow \therefore$ False

For what value of c does this sequence convergent

$$a_n = \frac{1}{n^c} \sin\left(\frac{\pi}{n}\right)$$

$$\lim_{n \rightarrow \infty} \frac{1}{n^c} \sin\left(\frac{\pi}{n}\right)$$

$$\lim_{n \rightarrow \infty} \frac{1}{n^c} \frac{\sin\left(\frac{\pi}{n}\right)}{\frac{\pi}{n}} \quad \frac{\pi}{n}$$

$$\lim_{n \rightarrow \infty} \frac{\pi}{n^c n} \frac{\sin\left(\frac{\pi}{n}\right)}{\frac{\pi}{n}}$$

$$\lim_{n \rightarrow \infty} \frac{\pi}{n^{c+1}} \frac{\sin\left(\frac{\pi}{n}\right)}{\frac{\pi}{n}} \quad (1)$$

لأن $\frac{\sin\left(\frac{\pi}{n}\right)}{\frac{\pi}{n}} \rightarrow 1$ $\therefore$ $\lim_{n \rightarrow \infty} \frac{\pi}{n^{c+1}}$ $\rightarrow 0$ $\therefore$ $a_n \rightarrow 0$
 $\oplus$ أو ∞

$\Rightarrow a_n$ is Convergent when
 $c+1 > 0$
 $c > -1$
 $c \in [-1, \infty)$

Find a formula for the general term a_n of the sequence , assuming that the pattern of the first terms continuous.

Then determine whether the sequence is converges or diverges. If converges find the limit.

$$\left\{ 0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots \right\}$$

$$a_n = \frac{n-1}{n}$$

$$\lim_{n \rightarrow \infty} \frac{n-1}{n}$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{n} [1 - \frac{1}{\cancel{n}}]}{\cancel{n}} = 1 - 0 = 1$$

$\therefore a_n$ is convergent

Find a formula for the general term a_n of the sequence , assuming that the pattern of the first terms continuous.

Then determine whether the sequence is converges or diverges. If converges find the limit.

$$\{ 5, 5, 5, 5, 5, 5, 5, \dots \}$$

$$a_n = 5$$

$$\lim_{n \rightarrow \infty} 5 = 5$$

$\therefore a_n$ is convergent



Text book questions



Find a formula for the general term a_n of the sequence , assuming that the pattern of the first terms continuous.

$$\left\{ \frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}, \frac{1}{10}, \dots \right\}$$

$$a_n = \frac{1}{2n}$$



Extra

Determine whether the sequence is converges or diverges. If converges find the limit.

$$\lim_{n \rightarrow \infty} \frac{1}{2n} = \frac{1}{\infty} = 0 \rightarrow \therefore a_n \text{ is Convergent}$$

Find a formula for the general term a_n of the sequence , assuming that the pattern of the first terms continuous.

$$\left\{ 1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \frac{1}{9}, \dots \right\}$$

$$a_n = \frac{1}{2n-1}$$



Extra

Determine whether the sequence is converges or diverges. If converges find the limit.

$$\lim_{n \rightarrow \infty} \frac{1}{2n-1} = \frac{1}{\infty} = 0 \rightarrow \therefore a_n \text{ is Convergent}$$

$$\text{Let } \{a_n\} = \left\{ 0, \frac{\ln 2}{4}, \frac{\ln 3}{9}, \frac{\ln 4}{16}, \frac{\ln 5}{25}, \dots \right\}$$

(a) Does $\{a_n\}$ converge or diverge?

(b) Find a_{100}

$$a_n = \left[\underset{\textcircled{1}}{0}, \underset{\textcircled{2}}{\frac{\ln 2}{4}}, \underset{\textcircled{3}}{\frac{\ln 3}{9}}, \underset{\textcircled{4}}{\frac{\ln 4}{16}}, \underset{\textcircled{5}}{\frac{\ln 5}{25}}, \dots \right]$$

$$a_n = \frac{\ln n}{n^2}$$

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n^2} \quad \frac{\infty}{\infty}$$

$$\text{Let } f(x) = \frac{\ln x}{x^2}$$

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x^2}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{2x} = \lim_{x \rightarrow \infty} \frac{1}{2x^2} = \frac{1}{\infty} = 0$$

$\Rightarrow \therefore a_n$ is convergent to 0

$$(b) a_{100} = \frac{\ln 100}{(100)^2}$$



Text book questions



Find a formula for the general term a_n of the sequence, assuming that the pattern of the first terms continuous.

Then determine whether the sequence is converges or diverges. If converges find the limit.

$$\left\{ -3, 2, -\frac{4}{3}, \frac{8}{9}, -\frac{16}{27}, \dots \right\}$$

$$a_n = -3 \left(-\frac{2}{3} \right)^{n-1}$$

$$\lim_{n \rightarrow \infty} -3 \left(-\frac{2}{3} \right)^{n-1} = 0 \quad \text{since } \left| -\frac{2}{3} \right| < 1$$

$\therefore a_n$ is convergent.

Determine whether the sequence is converges or diverges. If converges find the limit.

$$\{ 0, 1, 0, 0, 1, 0, 0, 0, 1, \dots \}$$

The limit of the sequence **DNE**

$\therefore$ The sequence is divergent.



Multiple choice



The general term of the sequence

Fall 2018

$\left\{-1, \frac{2}{9}, \frac{-4}{25}, \frac{8}{49}, \dots\right\}$ is : for $n = 0, 1, 2, \dots$

a) $a_n = \frac{(-1)^n 2^{n+1}}{(2n+1)^2}$. b) $a_n = \frac{(-1)^{n+1} 2^n}{(2n+1)^2}$

c) $a_n = \frac{(-1)^n 2^n}{(2n+1)^2}$ d) $a_n = \frac{2^n}{(2n+1)^2}$

e) $a_n = \frac{(-1)^{n+1} 2^n}{(2n+3)^2}$.

(b) ✓



Multiple choice



$$\text{Let } \{a_n\}_{n=1}^{\infty} = \left\{ -\frac{1}{2}, \frac{8}{3}, -\frac{27}{4}, \frac{64}{5}, -\frac{125}{6}, \dots \right\}$$

Assuming that the pattern continuous, a formula for general term a_n is

a) $(-1)^n \frac{n}{2^n}$ b) $(-1)^n \frac{8^{n-1}}{n+1}$

c) $(-1)^n \frac{3^n}{n+1}$ d) $a_n = (-1)^n \frac{n^3}{n+1}$

Dec 2017

a) $n=4 \longrightarrow a_4 = \frac{4}{2^4}$ X

b) $n=4 \longrightarrow a_4 = \frac{8^3}{5}$ X

c) $n=4 \longrightarrow a_4 = \frac{3^4}{5} = \frac{81}{5}$ X

d) $n=4 \longrightarrow a_4 = \frac{4^3}{5} = \frac{64}{5}$ ✓

(d) ✓